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VRFlipAuth: Usable and Secure Authentication based on Flip Game in Virtual Reality

Zibo Zheng* Jinan University
BoYu Gao† Jinan University
Huawei Tu‡ La Trobe University
Hai-Ning Liang§ HKUST (Guangzhou)
Tingjie Wan¶ Jinan University
Jian Weng‖ Jinan University

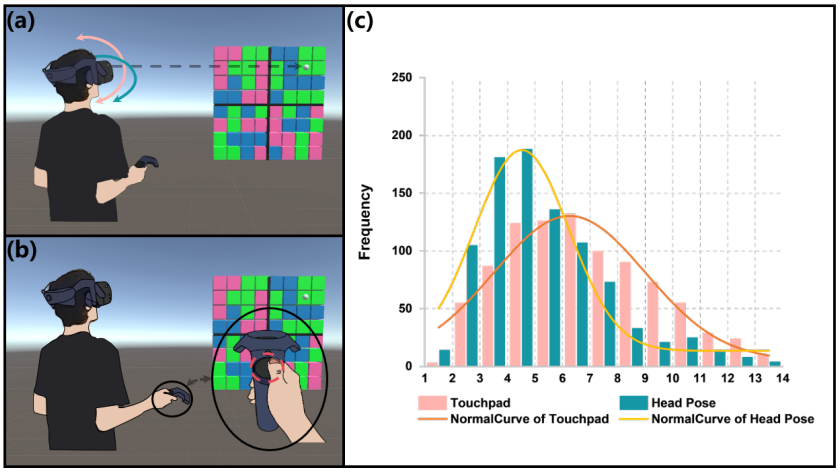

Figure 1: The concept of VRFlipAuth. a. using head pose to select, controlling the indicator by moving the head up, down, left, and right. b. Using the touchpad to select, and the thumb to control the touchpad to the indicator. The direction of thumb movement is mapped to the indicator via the touchpad. c. Frequency of authentication time in two input methods. Histograms showed that the authentication times were not normally distributed, and manifest a positive skewness. the input method had statistical significance on the authentication time. Using the head pose was significantly faster than that using the touchpad on the controller.

Abstract

With the growing use of Virtual Reality (VR) in various applications, authentication is crucial, especially in immersive VR environments. We propose VRFlipAuth, a novel knowledge-based authentication scheme inspired by the flip game, using spatially located cubes as a password. The pilot study demonstrates that VRFlipAuth provides fast authentication, offering an initial solution for authentication in immersive VR environments.

Index Terms: Virtual Reality, flip game, knowledge-based authentication.

1 Introduction

Virtual Reality (VR) is increasingly used in gaming, education, and collaborative platforms, providing immersive and engaging experiences. However, this immersion often reduces users' awareness of security risks, leaving them vulnerable to information leaks and observation-based attacks [1]. As VR platforms evolve into social spaces, secure and usable authentication mechanisms are urgently needed. Existing VR authentication methods mainly fall into two categories: biometric-based and knowledge-based [5, 4]. Biometric methods, like facial recognition or gait analysis, offer fast and unique authentication but rely on specialized hardware and raise ethical concerns [2, 7]. In contrast, knowledge-based methods, such as PINs and graphical passwords, are easier to implement on current VR devices and can leverage 3D interaction features like head pose and gestures [3]. However, the existing knowledge-based authentication methods are vulnerable to man-in-the-middle (MITM) attacks. In this paper, we propose VRFlipAuth, a knowledge-based authentication system inspired by flip games. It encodes passwords through cube-flipping operations on a flat plane, dynamically updating the password and improving the security.

*e-mail: qitiantaile@stu2024.jnu.edu.cn
†e-mail: bygao@jnu.edu.cn (Corresponding author)
‡e-mail: h.tu@latrobe.edu.au
§e-mail: hainingliang@hkust-gz.edu.cn
¶e-mail: tjwan@jnu.edu.cn
‖e-mail: cryptjweng@gmail.com

2 VRFlipAuth Design

We propose VRFlipAuth (Fig.1), a VR authentication solution inspired by the Flip Game. In this game, players flip squares to change their colors. Similarly, in VRFlipAuth, users select passwords by flipping cubes on a plane, which serves as a means of verifying their identity.

2.1 Layout Design

The layout of VRFlipAuth uses a 2D plane for password input, offering two main advantages [6]: **1)** Smooth, continuous input without depth changes, reducing operation time; **2)** Faster, more accurate selections with lower error rates compared to 3D layouts. The plane is composed of an 8x8 cube array. Each cube uses red, blue, and green textures, and the system supports color-blind users.

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2.2 Input Selection

Based on the requirement of the flip game, the main interactions in VRFlipAuth are to flip the cube at specific positions within the plane that users have predefined as the password. In this study, we select head pose and touchpad in the controller interaction [4, 3] (Fig.1). The sliding of the head enables the cursor to move on the plane, while using the touchpad on the controller also supports it. While confirmation occurs at a particular position, the cubes will be flipped immediately.

3 User Study

The user study aimed to evaluate the usability, memorability, and security of VRFlipAuth in a simplified yet integrated experiment. Specifically, we aim to understand the balance between usability and security of this proposed method.

3.1 Participants and Apparatus

We recruited 24 participants (12 women and 12 men) aged between 18 and 30 years from the local campus, and no color-blind participants. Ten participants had used VR devices before this experiment, and we assigned them to two groups for two interaction methods, respectively. All participants used HTC VIVE Pro2 (4986 px x 2448 px) for all experiments, and the SteamVR Plugin for Unity 3D for the HTC VIVE controller communication.

3.2 Procedure

The study was conducted in two phases. First, participants were introduced to VRFlipAuth and practiced using two input methods (head pose and touchpad). After training, they performed authentication tasks: entering predefined 4-symbol passwords with different spatial distributions on an 8x8 plane. We recorded authentication time, entry accuracy, and collected workload ratings (NASA-TLX) and subjective feedback. Second, after three days, participants returned and attempted to re-enter three self-set passwords (weak, medium, strong), assessing memorability. They used their preferred input method and had three attempts per password, allowing us to evaluate short-term memorability and explore users' self-chosen password strategies.

3.3 Initial Results

The study demonstrates that VRFlipAuth effectively balances usability and memorability. Participants were able to authenticate quickly and accurately, and recall their passwords after a short delay.

The two-way repeated measures ANOVA with Greenhouse Geisser correction showed that the input method ($F_{1,458} = 56.611, p < 0.001, n_p^2 = 0.110$) resulted in a statistically significant difference in the authentication time. Participants achieved fast and accurate authentication. Further analysis was performed to distinguish the impact of the independent variable. Bonferroni-corrected post hoc showed that the authentication time using the head pose ($M = 5.271s, SD = 0.132s$) was significantly faster ($p < 0.001$) than that using the touchpad on the controller ($M = 6.672s, SD = 0.132s$). The average entry accuracy in all situations was 97.72%, though accuracy did not differ significantly ($p = 0.052 > 0.05$). NASA-TLX scores indicated low to moderate workload, with head pose ($M = 39.63, SD = 14.33$) rated easier and faster than touchpad input ($M = 42.89, SD = 14.63$). Participants reported feeling that the dynamic flipping increased security, especially against Shoulder Surfing.

For memorability, after three days, participants recalled 100% of weak, 81.8% of medium, and 72.7% of strong self-set passwords. Many participants reported using spatial grouping, symmetry, and personal cues to make passwords easier to remember.

For attack time, the input methods ($F_{1,176} = 0.920, p = 0.339 > 0.05, n_p^2 = 0.005$) had no significant impact on the attack time. However, the attack models ($F_{1,334} = 5.374, p = 0.021 < 0.05, n_p^2 = 0.021$) had a significant impact on the attack time. MITM attackers ($M = 119.324s, SD = 45.76s$) consumed more time than Shoulder surfing attackers ($M = 106.804s, SD = 41.91s$). Shoulder Surfing attacks achieved only 1.81% success, and MITM attacks achieved 2.26%.

4 Discussion and Conclusion

Our user study shows that VRFlipAuth achieves a practical balance between usability and security in VR authentication. The dynamic flip-based interface enabled fast and accurate password entry while maintaining good memorability even for self-set strong passwords. The head pose input is faster and perceived as more intuitive than touchpad input, which demonstrates that VRFlipAuth supports efficient interactions without imposing high cognitive or physical workloads. The dynamic interface updates significantly complicate observation-based attacks, while the flip-based spatial encoding improves both usability and security. MITM attackers spent significantly more time, suggesting that VRFlipAuth's dynamic interface updates increase cognitive effort for observers, even with full VR access. This supports its potential as a pre-authentication mechanism for sensitive VR actions in social and collaborative environments. These results indicate that VRFlipAuth offers an initial practical, game-inspired, and secure authentication solution for VR. However, the study involved a small sample and a short-term evaluation. Future work will include a larger, diverse sample, extended usage in the realistic attack scenarios to validate and refine VRFlipAuth.

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References

[1] J. M. Jones, R. Duezguen, P. Mayer, M. Volkamer, and S. Das. A literature review on virtual reality authentication. In *Human Aspects of Information Security and Assurance: 15th IFIP WG 11.12 International Symposium, HAISA 2021, Virtual Event, July 7–9, 2021, Proceedings 15*, pp. 189–198. Springer, 2021. 1

[2] J. Liebers, P. Laskowski, F. Rademaker, L. Sabel, J. Hoppen, U. Gruenefeld, and S. Schneegass. Kinetic signatures: A systematic investigation of movement-based user identification in virtual reality. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems*, pp. 1–19, 2024. 1

[3] F. Mathis, J. H. Williamson, K. Vaniea, and M. Khamis. Fast and secure authentication in virtual reality using coordinated 3d manipulation and pointing. *ACM Transactions on Computer-Human Interaction (ToCHI)*, 28(1):1–44, 2021. 1, 2

[4] N. Noah and S. Das. From pins to gestures: Analyzing knowledge-based authentication schemes for augmented and virtual reality. *IEEE Transactions on Visualization and Computer Graphics*, 2025. 1, 2

[5] S. Stephenson, B. Pal, S. Fan, E. Fernandes, Y. Zhao, and R. Chatterjee. Sok: Authentication in augmented and virtual reality. In *2022 IEEE Symposium on Security and Privacy (SP)*, pp. 267–284. IEEE, 2022. 1

[6] R. Turkmen, C. Nwagu, P. Rawat, P. Riddle, K. Sunday, and M. B. Machuca. Put your glasses on: A voxel-based 3d authentication system in vr using eye-gaze. In *2023 IEEE conference on virtual reality and 3D user interfaces abstracts and workshops (VRW)*, pp. 947–948. IEEE, 2023. 1

[7] T. Wan, L. Zhang, Y. Xu, Z. Guo, B. Gao, and H.-N. Liang. Analysis and design of efficient authentication techniques for password entry with the qwerty keyboard for vr environments. *IEEE Transactions on Visualization and Computer Graphics*, 2024. 1

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